The Value and Practical Approaches of Artificial Intelligence in Enhancing English Teaching

Li Jiang[1] , Ruoyu Wang[2], Meng Wei[3], Wei Wang[4], Minzhen Du[5*]

1.College of Education, Zhejiang Normal University/ Zhengzhou No.36 Senior High School, China, jlzjnu36@yeah.net;

2.College of Education, Zhejiang Normal University/Diehu Primary School, Zhengzhou Jingkai District,China,18538108052@163.com

3.Songshan Road School of Erqi District, China,1163179108@qq.com;

4.College of Education, Zhejiang Normal University/Chaoyang Normal School Affiliated Primary School, China,wangweiblcu@163.com

5.College of Education, Zhejiang Normal University,China,315090494@qq.com

Abstract. Leveraging generative AI to address inherent challenges in China's English teaching and optimize the educational ecosystem has become a central focus of English education reform in the new era. This study follows a logical framework of problem identification, value clarification, and practical implementation, focusing on four core issues in Chinese English teaching: rigid instructional models, lack of personalized instruction, inefficient assessment systems, and suboptimal learning experiences. By examining real-world teaching scenarios, it analyzes the core value of generative AI in enhancing English instruction—including deep inspiration through conversational interaction, precise delivery of personalized learning supported by accurate data, efficient optimization of teaching processes via technological iteration, and scientific upgrades to assessment systems enabled by intelligent algorithms.

Keywords: Artificial Intelligence; Language Education; English Teaching

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[*] Corresponding Author:Minzhen Du (315090494@qq.com)

1. Introduction

Minister of Education Huai Jinpeng emphasized that artificial intelligence technology must be fully integrated into all stages and processes of 1education. This underscores the urgent need to combine AI with various aspects of teaching[1]. Artificial intelligence mimics human intelligence, aiming to endow computer systems with human-like cognitive abilities and enable them to perform tasks similar to those performed by humans. The technology integrates diverse methodologies such as machine learning, deep learning, natural language processing, computer vision, expert systems, and intelligent agents[2]. By deeply integrating generative AI into 2the entire English teaching process—including instruction, learning, assessment, and testing—it enhances students' learning efficiency and language proficiency while preserving the strengths of traditional pedagogy. It promotes personalized education, provides timely feedback, assists teachers in better understanding student progress, and optimizes instructional workflows[3]. After examining challenges in English teaching, this study analyzes the potential of AI from an educational perspective and offers recommendations for educators to adapt to the digital age, achieve substantive improvements in English instruction, help students alleviate learning pressures, enjoy the joy of learning, and achieve rapid progress[4].

2. The Current Practical Challenges in English Teaching

2.1. Rigid Teaching Models: The Advantages and Practical Limitations of Traditional Methods

As a classic approach in China's English education, the grammar translation method holds irreplaceable value in systematically constructing language knowledge frameworks, precisely imparting grammatical rules, and solidifying vocabulary foundations—serving as a crucial basis for students to grasp the fundamental logic of the English language. However, under contemporary language teaching requirements, its practical limitations have become increasingly apparent[5]. First, the predominantly one-way instructional approach neglects the communicative essence of language, failing to cultivate students' practical language application skills. Second, in large-class settings, teachers struggle to meet the diverse learning needs of students at different proficiency levels, resulting in advanced learners being under-challenged while struggling learners fall behind. Third, the teaching context remains disconnected from real-life contexts, depriving students of authentic language practice opportunities and predisposing them to "mute English" (English without practical use)[6]. Clearly, these inherent flaws in traditional teaching methods have long undermined the overall effectiveness of English education in China, preventing optimal educational outcomes. This issue stems not from the method itself but rather from insufficient technological empowerment and innovative teaching methodologies that fail to align with modern pedagogical demands for personalized, application-oriented English instruction[7].

2.2. Inefficient Learning Experience: Students' Core Learning Barriers and Lack of Intrinsic Motivation

Currently, most Chinese students still exhibit weak foundations in English and have not established a solid knowledge system. The root cause of this issue lies not in students' inherent limitations, but rather in their lack of scientific learning methods and sustained intrinsic motivation. Specifically: First, many students demonstrate significant anxiety toward English learning, lacking initiative and enduring interest-driven motivation. These students generally possess inadequate self-directed learning skills and self-regulation awareness, making it difficult for them to fully engage their proactive efforts. Their low participation levels both in class and after lessons lead to passive knowledge absorption, poor learning outcomes, and a vicious cycle of declining confidence. Second, most students remain dependent on teachers assigning tasks and checking homework, lacking the ability to independently develop study plans, refine learning strategies, or reflect on learning challenges—making it hard for them to meet autonomous and personalized learning needs. Third, the negative transfer effect of native language thinking is pronounced; students often succumb to Chinese cognitive patterns and expression habits in English communication, resulting in widespread "Chinglish." The absence of timely correction and guidance exacerbates this issue, compromising the standardization of their linguistic expression. Additionally, the lack of authentic language interaction scenarios prevents students from practicing and reinforcing acquired knowledge, leading to monotonous learning experiences that fail to sustain lasting interest. Overall, due to insufficient foundational competence, they frequently struggle to independently complete various English learning tasks[8].

2.3. Shortcomings of the Evaluation System: The Subjectivity and Inefficiency of Traditional Evaluation Methods

In the closed-loop teaching-learning-assessment system of English instruction, the evaluation phase serves as a critical mechanism for assessing instructional effectiveness and guiding student learning. However, traditional English assessment methods exhibit significant limitations. First, oral assessment is highly subjective and costly to implement; it predominantly relies on manual grading by teachers, which is susceptible to subjective factors such as instructors 'professional expertise, emotional states, and evaluation criteria. Large-scale oral assessments also require substantial staffing and time investments, making routine, standardized implementation challenging. Second, grading objective question types is inefficient—teachers spend considerable time correcting multiple-choice and fill-in-the-blank exercises, compromising time allocated for lesson preparation, pedagogical research, and personalized tutoring. Third, subjective assessment lacks precision; traditional essay and translation evaluations primarily involve overall scoring, failing to provide targeted feedback on specific student issues (e.g., grammatical errors, logical inconsistencies, or vocabulary deficiencies), hindering students from accurately identifying their weaknesses. Fourth, assessments are delayed—they mainly consist of unit tests, midterm exams—which make it difficult to monitor students' learning progress in real-time or promptly identify and address temporary learning obstacles[9].

2.4. Absence of Personalized Teaching: Resource Limitations and Competency Bottlenecks in Traditional Instruction

Personalized teaching is one of the core objectives of English education, yet its implementation is constrained by the resource and capability limitations inherent in traditional teaching approaches. Firstly, there is a shortage of teaching staff; individual instructors struggle to meet the diverse learning needs of dozens of students and cannot develop tailored learning plans or provide targeted guidance for each student. Secondly, there is a lack of precise student data support; teachers find it difficult to comprehensively and systematically track students 'learning progress, knowledge acquisition, and study habits, relying instead on experience to assess their learning needs, which results in unscientific instructional decisions. Thirdly, teaching resources are highly homogeneous; traditional resources primarily consist of standardized textbooks and course materials that fail to be personalized according to students' foundational knowledge, interests, and learning goals, thus failing to address the varied learning requirements of different students.

3. The Value and Significance of Artificial Intelligence in English Teaching

3.1. Model Innovation: Empowering the Transformation and Upgrading of Traditional Teaching to Optimize the Educational Ecosystem

Conversational AI, with its capabilities in natural language processing, generative creation, and multimodal interaction, does not negate traditional teaching models but rather provides technological empowerment for their transformation and upgrading, optimizing the educational ecosystem. On one hand, teachers need to continuously seek innovative teaching methods and inspiration when designing lesson plans and conducting daily instructional activities. The real-time responsiveness of large AI models can stimulate teachers' teaching creativity while serving as personalized intelligent advisors and auxiliary tools for each educator. Taking English teaching as an example, this technology assists teachers in generating instructional design ideas and activity proposals during course preparation, while also aiding in the creation of teaching materials. Teachers can leverage these AI models to develop creative concepts around specific teaching themes, enabling them to design more systematic, comprehensive, and scientifically sound lesson plans. For instance, when the author entered the English prompt "Make a lesson outline with learning objectives as well as creative activities" into the iFlytek Spark large model, the system immediately generated corresponding teaching objectives, activity designs, and expected outcomes for a travel-themed lesson, as illustrated in Figure 1.

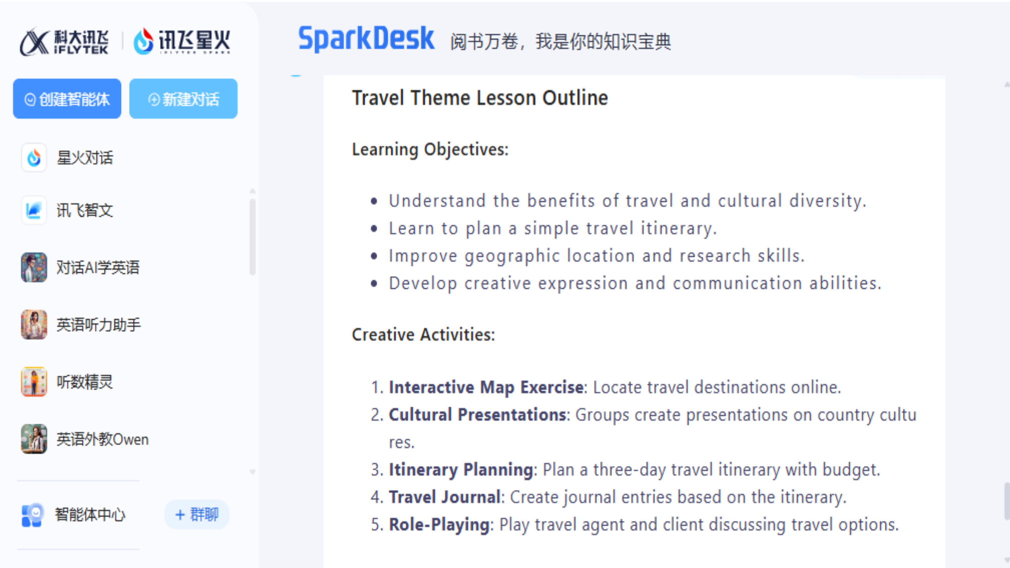

Figure 1: Example of instructional design generated by iFlytek Spark

However, this instructional design cannot be directly applied to classroom teaching. Its framework possesses basic feasibility for implementation, and the activity design covers dimensions such as language application and practical exploration. Nevertheless, the task difficulty should be adjusted according to students 'actual foundational knowledge—for example, incorporating professional-related elements like overseas academic exchanges or cross-border study tours into itinerary planning. Moreover, the original teaching objectives did not integrate ideological and political education elements. Tasks related to introducing China's cultural and tourism resources could be added during the cultural presentation phase, guiding students to tell China's stories effectively in English and fostering cultural confidence through cross-cultural comparisons.

On the other hand, artificial intelligence enables innovation in teaching methodologies by transforming traditional one-way instruction into diverse interactive formats involving teacher-student, peer-to-peer, and human-machine interactions. By simulating authentic language communication scenarios, it allows students to apply linguistic knowledge in practice, addressing the disconnect between conventional classroom [3] settings and real-life contexts. Notably, AI-generated instructional designs are not mere copy-pastes but serve as creative references for educators, who must optimize and adapt them based on learners' specific circumstances and teaching objectives to enhance their relevance.

3.2. Experience Upgrade: Precisely Addressing Students' Learning Barriers and Stimulating Their Intrinsic Motivation

Through instant precise feedback, personalized learning path planning, and immersive learning environments, artificial intelligence effectively addresses students 'core learning challenges, enhances the learning experience, and stimulates their intrinsic motivation. Firstly, AI serves as a dedicated learning coach for students, interacting with them via voice or text to provide immediate, specific, and actionable corrections for language errors (e.g., pronunciation, grammar, vocabulary), overcoming the limitation of traditional teaching where instructors cannot offer timely individual guidance. For instance, students can use Doubao's foreign teacher voice call feature to practice oral skills in realistic scenarios like restaurant ordering, hotel reservations, or street navigation—AI continuously corrects pronunciation and intonation while offering authentic expression suggestions, helping students clearly identify their weaknesses. Secondly, AI alleviates anxiety during oral practice; in stress-free learning environments, students avoid negative teacher evaluations or peer ridicule, confidently express themselves, and gradually improve their speaking proficiency and confidence. Finally, AI tailors personalized learning paths based on students' foundational levels, progress, and habits, delivering targeted resources that enable achievement at achievable difficulty levels. This approach fosters sustained interest and transforms passive learning into active engagement.

3.3. Efficiency Improvement: Comprehensive tracking of learning data throughout the entire process enables scientific teaching decision-making

The big data analytics, learning behavior tracking, and visualization capabilities of artificial intelligence provide the core foundation for personalized teaching, fundamentally addressing the lack of precise data support in traditional education and enabling scientific instructional decision-making. The intelligent teaching platform serves as a "third eye" for classroom instruction, meticulously recording comprehensive data—including teacher-student interactions, students 'learning progress, knowledge mastery, study habits, and error patterns—and automatically generating statistical reports and visual analytics. By analyzing this data, educators gain deep insights into each student's learning patterns, knowledge gaps, and individual characteristics, allowing them to develop tailored learning plans for personalized instruction. Simultaneously, teachers can adjust teaching pacing, optimize content, and refine methodologies based on collective class data, enhancing the scientific rigor and relevance of instructional decisions. Leveraging cutting-edge AI technologies such as cloud computing, machine learning, and speech recognition, English teachers can systematically map learning trajectories using analytical data and provide timely interventions to guide student learning behaviors, offering robust support for teaching strategies. The application of intelligent teaching technologies not only significantly enhances teachers 'data analysis capabilities but also enables deeper understanding of learners' dynamic progress and emotional fluctuations, delivering timely and precise support to ultimately achieve efficient and personalized educational outcomes. The implementation of such intelligent teaching platforms makes the entire instructional process more scientific and precise, significantly enhancing teaching effectiveness while improving students' learning experience.

3.4. Evaluation Optimization: Promoting the automation and scientific rigor of the evaluation system to achieve a closed-loop teaching-learning-evaluation cycle

Through automated assessment, objective scoring, precise feedback, and standardized evaluation practices, artificial intelligence technology drives the transformation and upgrading of English assessment systems, addressing the limitations of traditional methods and establishing a closed-loop cycle of teaching, learning, and evaluation. First, it enhances the precision of diagnostic assessments: By leveraging AI technologies, teachers can conduct accurate English proficiency evaluations before students begin their learning journey, enabling precise measurement of language abilities. This approach not only provides robust support for teachers to swiftly understand students 'cognitive and linguistic skills but also facilitates more refined and personalized instructional design, significantly improving teaching efficiency. Second, it automates the assessment process: The intelligent English listening and speaking system built on speech recognition technology fully automatizes oral test procedures and scoring, ensuring objectivity and accuracy while reducing implementation costs and making oral assessments routine. Third, it optimizes homework grading: AI systems efficiently grade multiple-choice and fill-in-the-blank questions, while providing comprehensive smart evaluations for subjective tasks like essays and translations—covering grammar, vocabulary, logic, and expression—with targeted correction feedback and improvement suggestions, helping students pinpoint specific areas for improvement. Fourth, it alleviates teachers' workload: AI frees educators from tedious repetitive grading and assessment tasks, freeing up time and energy for lesson planning, pedagogical research, and personalized tutoring, thereby enhancing overall teaching quality.

4. Practical Approaches to Empowering English Teaching with Artificial Intelligence

4.1. Conceptual Reconstruction: Establishing a Human-Machine Collaborative Teaching Philosophy and Adopting a Dialectical Approach to the Relationship Between Technology and Tradition

In the era of generative artificial intelligence, English teachers must first restructure their teaching philosophies by embracing a new collaborative human-machine approach and maintaining a balanced perspective on the relationship between AI technology and traditional pedagogy. On one hand, educators should abandon both the extremes of technological omnipotence and technological uselessness, recognizing that AI serves as an empowerment tool rather than a replacement for teaching. Its core value lies in addressing shortcomings of conventional instruction and optimizing the educational ecosystem—areas where human guidance, emotional engagement, and value cultivation remain irreplaceable. On the other hand, teachers need to actively transform their traditional mindset, evolving from mere knowledge transmitters into learning facilitators, instructional designers, and coordinators of human-machine collaboration. They should guide students toward adopting appropriate collaborative learning concepts, cultivate their ability to use AI for self-directed learning and problem-solving, while preventing excessive reliance on artificial intelligence that could undermine independent thinking skills. Furthermore, teachers must systematically enhance their digital literacy—not only mastering basic AI tool operations but also developing capabilities in designing AI-powered teaching applications, evaluating AI-generated content, analyzing instructional data, and fully exploring AI's potential in foreign language education to achieve deep integration of technology with teaching practices.

4.2. Tool Adaptation: Precisely select AI tools to achieve deep alignment between tools and teaching scenarios

When integrating large language models such as iFlytek Spark into language education practices, teachers must first keep pace with contemporary trends and thoroughly evaluate whether the selected models and tools truly align with their specific English teaching objectives. Different model types are typically suited to distinct language learning and instructional scenarios: conversational AI systems like iFlytek Spark and DouBao excel in lesson design, oral practice, and personalized tutoring; generative [4]AI platforms such as ChatGPT and Wenxin Yiyuan are ideal for creating teaching materials, grading essays, and translation exercises; while intelligent speech AI solutions from companies like iFlytek and NetEase Youdao are particularly effective for oral assessment and pronunciation correction. Additionally, educators should carefully regulate the duration of using these software tools and large language models in English instruction—overreliance may undermine teachers' critical thinking skills and motivation [5] for continuous learning. In its report "Artificial Intelligence in Education: Challenges and Opportunities for Sustainable Development," UNESCO warns that advancing AI technology risks leading learners into superficial thinking patterns, non-systematic learning approaches, potentially narrowing their cognitive horizons, hindering problem-solving abilities, and stifling higher-order thinking development. Therefore, English teachers should embrace a forward-looking approach, leveraging AI technologies to enhance rather than replace traditional teaching practices. Meanwhile, educators should proactively explore innovative teaching approaches closely integrated with artificial intelligence technologies to support and enhance the development of students' higher-order thinking skills.

4.3. Content Verification: A three-tier verification mechanism for AI-generated content ensures the accuracy of teaching materials.

In practical applications, artificial intelligence systems may occasionally make erroneous judgments or exhibit interpretive biases, particularly when handling complex or sensitive topics. These systems might choose to avoid answering questions or provide inappropriate conclusions, potentially leading to misunderstandings or misguidance. Therefore, educators must establish a three-tier verification mechanism for AI-generated content, integrating its assessment and validation as essential components of teaching practices to ensure content accuracy and relevance. The first tier involves professional knowledge verification: teachers utilize their English expertise to review AI-generated instructional designs, materials, and explanations, identifying errors in grammar, vocabulary, and cultural contexts. The second tier assesses instructional suitability by evaluating whether the content aligns with students' actual needs, specific teaching objectives, and national curriculum standards. The third tier examines value orientation, detecting potential negative influences, cultural biases, or erroneous viewpoints to ensure compliance with the fundamental mission of fostering moral integrity through education. Furthermore, teachers should transform this verification process into teaching resources, encouraging student participation in analyzing and discussing AI-generated content to cultivate critical thinking and information discernment skills. Through continuous monitoring and refinement, educators can effectively leverage AI technologies to create a more precise and secure learning environment for students.

4.4. Examining Challenges and Addressing Ethical Issues Arising from the Application of Artificial Intelligence Technologies

Integrating artificial intelligence (AI) technology into education may raise a range of ethical concerns, particularly in academic research, focusing on issues such as privacy protection, data security, and potential gaps in emotional engagement. The practical implementation of AI often requires collecting students 'personal information—including chat records—which poses risks to their privacy. For instance, in April 2023, Spain's Data Protection Agency (AEPD), following an Italian investigation, formally launched an inquiry into OpenAI. The investigation addressed three key issues: first, ChatGPT's large-scale collection and processing of personal data without explicit user consent, suspected violations of the EU's General Data Protection Regulation (GDPR); second, the platform's vulnerability to data breaches (OpenAI had previously experienced leaks of user conversations and payment information); third, the absence of effective age verification mechanisms for minors, potentially infringing on their data rights. Teaching is inherently an interactive process involving deep emotional connections between educators and students. However, AI-assisted instruction [6] may reduce such emotional exchanges, fostering a sense of detachment. A study on AI ethics in higher education found that students' preference for direct AI assistance—driven by its convenience—may undermine the quality of interpersonal interactions and emotional bonds among teachers, peers, and learners.

Therefore, when addressing these ethical issues, teachers must comprehensively assess the challenges involved and conduct thorough evaluations and planning across multiple dimensions—including legal frameworks, privacy protection, and data security—to minimize their negative impacts. Simultaneously, teachers should proactively guide students to fully understand the various risks associated with the practical use of artificial intelligence, thereby providing robust safeguards for their holistic development.

5. Conclusion

Generative artificial intelligence offers groundbreaking technological empowerment for the reform and development of English teaching. Its core value lies not in negating or replacing traditional English instruction, but in addressing four fundamental challenges in China's English education—rigid teaching models, inefficient learning experiences, deficiencies in assessment systems, [7]and the absence of personalized teaching—while preserving the strengths [8]of traditional pedagogy in knowledge foundation and humanistic guidance. This approach drives the transformation of English teaching from knowledge transmission to competency cultivation, from one-way indoctrination to human-machine collaboration, and from uniform instruction to personalized learning. As educational intelligence continues to advance, AI will play a pivotal guiding role. Currently, AI-enhanced English teaching remains in its early developmental [9] stage, requiring further empirical research to explore more targeted human-machine collaborative teaching models. Concurrently, it is essential to strengthen teachers' digital literacy, refine ethical guidelines and management frameworks for AI applications in education, and ensure AI truly serves as a catalyst for English teaching reform—enhancing overall teaching quality while fostering students' core English competencies and cross-cultural communication skills. By leveraging modern information technologies like AI, educators can innovate teaching methodologies, accelerate educational reforms, and ultimately elevate the overall standard of English instruction.

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* Corresponding Author:Minzhen Du (315090494@qq.com)